State Estimation and Filtering for Robotic Perception

AA273

Linear-Gaussian Model Markovianity, Cond. Indept Measurements,
Kalman Filter MSE Optimality, Steady State KF, Extended Kalman Filter



Stanford University

MSL

AA273: State Estimation and Filtering for Robotic Perception

Announcements:

- HW 4 due Fri Feb 13
- Group signup sheet posted (groups from 1 – 4 people)
- Midterm: Feb 5-7
  - 5 hour timed take home exam, you choose any 5 hours in 48 hour window
  - Open book, notes, lecture videos
  - No internet, gen AI, or collaboration, coding assistants are allowed.
- Project Proposal: due Tue Feb 17
  - 2-3 pages, Latex
  - Related work >= 10 refs, ~1 page
  - Proposed work ~ 1 page
  - Bibtex refs

Lecture 9:

- Linear-Gaussian SS Model Markoviantiy and Linearly Indept. Measurements
- Kalman Filter MSE optimality
- Steady State KF
- Extended Kalman Filter (EKF)

Recall: L-G SS Model and KF Eqns

Dynamics eqn.:

$$X_{t+1} = A_t X_t + B_t u_t + W_t$$

Measurement eqn.:

$$Y_t = C_t X_t + D_t u_t + V_t$$

Initial condition:

$$X_0 \sim \mathcal{N}(\mu_{0|0}, \Sigma_{0|0})$$

Whiteness and Indep.:

$$W_t \sim GWN(0, Q_t)$$

$$V_t \sim GWN(0, R_t)$$

X_0, W_t, V_τ independent for all t, τ

Given: $\mu_{0|0}, \Sigma_{0|0},\ A_t, B_t, C_t,\ Q_t, R_t$

(Assume $D_t = 0$)

For: $t = 1, 2, \ldots$

Prediction:

$$\mu_{t|t-1} = A_{t-1}\mu_{t-1|t-1} + B_{t-1}u_{t-1}$$

$$\Sigma_{t|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1}$$

Update:

$$\mu_{t|t} = \mu_{t|t-1} + K_t(y_t - C_t\mu_{t|t-1})$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1}$$

$$K_t = \Sigma_{t|t-1}C_t^T(C_t\Sigma_{t|t-1}C_t^T + R_t)^{-1}$$

Loop

Linear Structure of Jointly Gaussian RVs

Theorem:

$$Z \sim \mathcal{N}(\mu, \Sigma)$$

If and only if for all $(X, Y) = Z$ with $X \sim \mathcal{N}(\mu_x, \Sigma_x)$, there exists a matrix M and RV $V \sim \mathcal{N}(\mu_v, \Sigma_v)$ independent of X such that

$$Y = MX + V$$

Intuition: Any components of a Gaussian RV can be written as a linear combination of the other components, plus Gaussian "noise" independent of those other components. I.e., All the information that some components have about the others is represented by a linear mapping.

E.g., Measurement Model

Recall the linear Gaussian measurement model:

$$Y = CX + Du + V$$

Where: $X \sim \mathcal{N}(\mu_x, \Sigma_x)$, $V \sim \mathcal{N}(0, R)$, and $Cov(X, V) = 0$

Re-write this as

$$Y = CX + \tilde{V} \qquad \tilde{V} \sim \mathcal{N}(Du, R) \text{ , and } Cov(X, \tilde{V}) = 0$$

This is **the only way** to ensure joint Gaussianity for (X, Y)

$$(X, Y) \sim \mathcal{N}(\mu, \Sigma)$$

E.g., Gaussian Estimation Equations

Recall the Gaussian estimation equations:

$$\mu_{x|y} = \mu_x + \Sigma_{xy}\Sigma_y^{-1}(y - \mu_y)$$

$$\Sigma_{x|y} = \Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx}$$

We also know from joint Gaussianity that:

$$X = MY + W \qquad W \sim \mathcal{N}(\mu_w, \Sigma_w)$$

What are M, (μ_w, Σ_w) ?

$$E[X \mid Y = y] = \mu_{x|y} = My + \mu_w$$

$$\mu_{x|y} = \Sigma_{xy}\Sigma_y^{-1}y + (\mu_x - \Sigma_{xy}\Sigma_y^{-1}\mu_y)$$

$$Cov(X \mid Y = y) = E[(My + W - (My + \mu_w))(My + W - (My + \mu_w))^T] = \Sigma_w$$

$$Cov(X \mid Y = y) = \Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx}$$

We have: $M = \Sigma_{xy}\Sigma_y^{-1}$, $\mu_w = (\mu_x - \Sigma_{xy}\Sigma_y^{-1}\mu_y)$, $\Sigma_w = \Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx}$

Conditionally Independent Measurements in Linear-Gaussian Model

Consider a jointly Gaussian sequence of state and measurements:

$$(X_{0:t}, Y_{1:t}) \sim \mathcal{N}(\mu, \Sigma)$$

We know that there exists matrix M_t and Gaussian noise $\tilde{V}_t$ such that:

$$Y_t = M_t \begin{bmatrix} X_{0:t} \\ Y_{1:t-1} \end{bmatrix} + \tilde{V}_t \qquad \begin{aligned} Cov(X_\tau, \tilde{V}_t) &= 0 \quad \forall \tau \le t \\ Cov(Y_\tau, \tilde{V}_t) &= 0 \quad \forall \tau < t \end{aligned}$$

Re-write as:

$$Y_t = \cancel{\Phi_{0:t-1} X_{0:t-1}} + C_t X_t + \cancel{\Psi_{1:t-1} Y_{0:t-1}} + \tilde{V}_t$$

Cond. Indep. Measurements: For Y_t indep. of $(X_{0:t-1}, Y_{1:t-1})$ given X_t we require $\Phi_{0:t-1} = 0$ and $\Psi_{1:t-1} = 0$

$Y_t = C_t X_t + \tilde{V}_t, \quad \tilde{V}_t \sim \mathcal{N}(\mu_{\tilde{v}_t}, R_t)$

Define: $\mu_{\tilde{v}_t} = D_t u_t + \mu_{v_t}$



$$\boxed{Y_t = C_t X_t + D_t u_t + V_t, \quad V_t \sim \mathcal{N}(\mu_{v_t}, R_t)}$$

Aside: recall whiteness of V_t ensures $Cov(X_\tau, \tilde{V}_t) = 0 \quad \forall \tau \le t$, $Cov(Y_\tau, \tilde{V}_t) = 0 \quad \forall \tau < t$.

Markovianity in Linear-Gaussian Model

Consider a jointly Gaussian sequence of state and measurements:

$$(X_{0:t+1}, Y_{1:t}) \sim \mathcal{N}(\mu, \Sigma)$$

We know that there exists matrix N_t and Gaussian noise $\tilde{W}_t$ such that:

$$X_{t+1} = N_t \begin{bmatrix} X_{0:t} \\ Y_{1:t} \end{bmatrix} + \tilde{W}_t \qquad \begin{aligned} Cov(X_\tau, \tilde{W}_t) &= 0 \quad \forall \tau \le t \\ Cov(Y_\tau, \tilde{W}_t) &= 0 \quad \forall \tau \le t \end{aligned}$$

Re-write as:

$$X_{t+1} = \cancel{\Theta_{0:t-1} X_{0:t-1}} + A_t X_t + \cancel{\Xi_{1:t} Y_{0:t}} + \tilde{W}_t$$

Markovianity: For X_{t+1} indep. of $(X_{0:t-1}, Y_{1:t})$ given X_t we require $\Theta_{0:t-1} = 0$ and $\Xi_{0:t} = 0$.

$$X_{t+1} = A_t X_t + \tilde{W}_t$$

Define: $\mu_{\tilde{w}_t} = B_t u_t + \mu_{w_t}$

$$\Rightarrow \quad \boxed{X_{t+1} = A_t X_t + B_t u_t + W_t, \quad W_t \sim \mathcal{N}(\mu_{w_t}, Q_t)}$$

Aside: recall whiteness of W_t ensures $Cov(X_\tau, \tilde{W}_t) = 0 \quad \forall \tau \le t$, $Cov(Y_\tau, \tilde{W}_t) = 0 \quad \forall \tau \le t$.

Linear-Gaussian Model

Summary: The Linear-Gaussian SS model is the only model that

(i) Is jointly Gaussian for $(X_{0:t}, Y_{1:t})$

(ii) Markov

(iii) Has conditionally independent measurements

Simplifying assumptions: We also typically assume

(i) Zero mean measurement noise $\mu_{v_t} = 0$

(ii) Zero mean process noise $\mu_{w_t} = 0$

(iii) $D_t = 0$

But we can re-formulate the KF to relax all of these as well.

Minimum Mean Square Error Optimality of Kalman Filter (any Bayesian Estimator)

On the board and lecture notes.

Steady State Kalman Filter

On the board and lecture notes.

Extended Kalman Filter

$$X_{t+1} = f(X_t, u_t) + W_t, \quad W_t \sim GWN(0, Q_t)$$

$$Y_t = g(X_t) + V_t, \quad V_t \sim GWN(0, R_t)$$

Where A_t and C_t Are Jacobian matrices.

Details on the board and lecture notes.

Given: $\mu_{0|0}$, $\Sigma_{0|0}$, $f(\cdot,\cdot)$, $g(\cdot)$, Q_t, R_t

For: $t = 1, 2, \ldots$

Prediction:

$$\mu_{t|t-1} = f(\mu_{t-1|t-1}, u_{t-1})$$

$$\Sigma_{t|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1}$$

Update:

$$\mu_{t|t} = \mu_{t|t-1} + K_t(y_t - g(\mu_{t|t-1}))$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1}$$

$$K_t = \Sigma_{t|t-1}C_t^T(C_t\Sigma_{t|t-1}C_t^T + R_t)^{-1}$$

Loop